Equipment Maintenance Log System

A full-stack platform for tracking equipment, rooms, and append-only maintenance history. Built with React and Laravel.



What is EMLS?

Designed for institutions, laboratories, and offices to monitor equipment status, track room assignments, and keep a trustworthy record of every maintenance action.

- Monitor equipment status
- Track rooms and in-charge faculty
- Keep append-only maintenance histories
- Control access with user roles



3 user roles



4 equipment statuses



5 maintenance actions

Key features

Role-based access

Admin, Technician, and Room In-Charge

Equipment tracking

Property numbers, categories, rooms, statuses

Search and filter

By name, property number, or room

Room tracking

Buildings, floors, and faculty in charge

Audit trail

Append-only logs with void-by-reason

Safe deletion

Items with history are marked Disposed

Who can do what



Admin (Custodian)

- Manage equipment, rooms, and users
- Record maintenance logs
- Void invalid entries



Technician

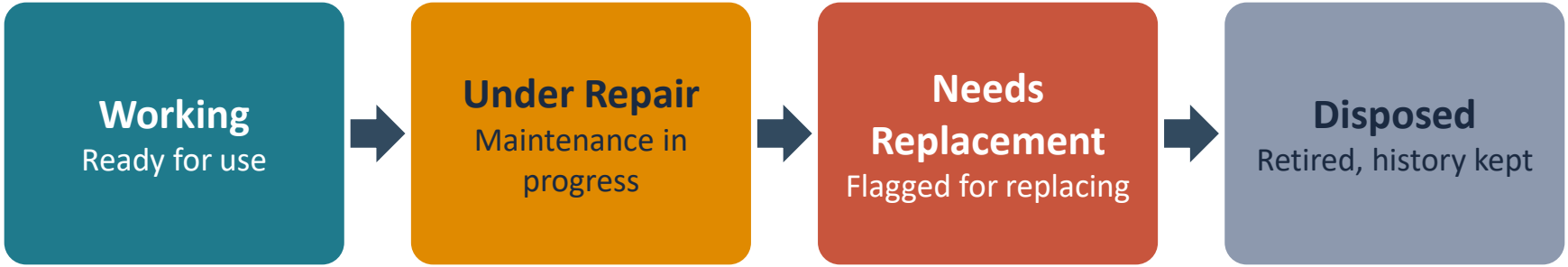
- View equipment
- Record maintenance activity logs



Room In-Charge (Faculty)

- View equipment in assigned rooms

Equipment status lifecycle



Safe deletion

Equipment with prior maintenance history is never erased. It is preserved and marked as Disposed.

Append-only audit trail

Maintenance actions

- Cleaned
- Repaired
- Modified
- Inspected
- Part Replaced

Void, never delete

1. An inaccurate log entry is found
2. Admin voids it and states a reason
3. The entry stays in the history, marked void

Technology stack



Frontend

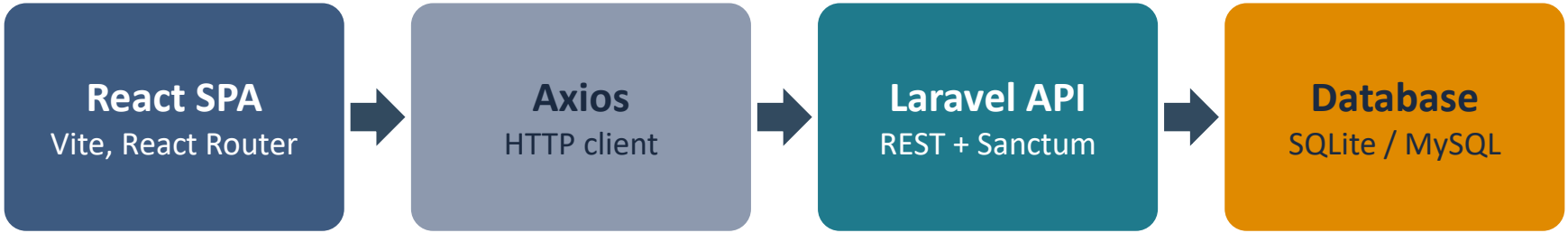
- React (Vite)
- React Router DOM
- Axios
- Modular CSS with design tokens, glassmorphism, and a dark theme



Backend

- Laravel (PHP 8.2+)
- Laravel Sanctum token authentication
- SQLite / MySQL
- REST API

System architecture



Controllers

Auth, Equipment, Log, Room, User

Models

Equipment, EquipmentLog, Room, User

Key API endpoints

Method	Endpoint	Purpose
POST	`/api/login`	Authenticate, receive Sanctum token
GET	`/api/me`	Authenticated user details
GET	`/api/equipment`	Paginated list (q, room_id, status)
POST	`/api/equipment`	Create equipment (Admin)
POST	`/api/logs`	Record log (Admin, Technician)
PATCH	`/api/logs/{log}/void`	Void a log with reason (Admin)
POST	`/api/users`	Register user with role (Admin)

Getting started

1. Backend (Laravel)

```
cd equipment-api
composer install
cp .env.example .env
php artisan key:generate
php artisan migrate:fresh --seed
php artisan serve --port=8000
```

2. Frontend (React)

```
cd frontend
npm install
npm run dev
```

Demo accounts (password: password)
admin@example.com | tech@example.com | faculty@example.com

Thank you

Tracked, auditable, role-aware equipment care. Open source under the MIT License.

github.com/amberroseperote62-cpu/Equipment-Maintenance-Log-System-React